



Wildland fire pollution monitoring in US EPA tests

CLIENT

US EPA

SECTOR

Research &
Consultancy

LOCATION

North Carolina, USA

PRODUCTS

Kunak AIR
Kunak AIR Cloud

Need

In 2017, the EPA and five other federal agencies issued a challenge to detect wildfires early using gas and particulate sensors to protect public health.

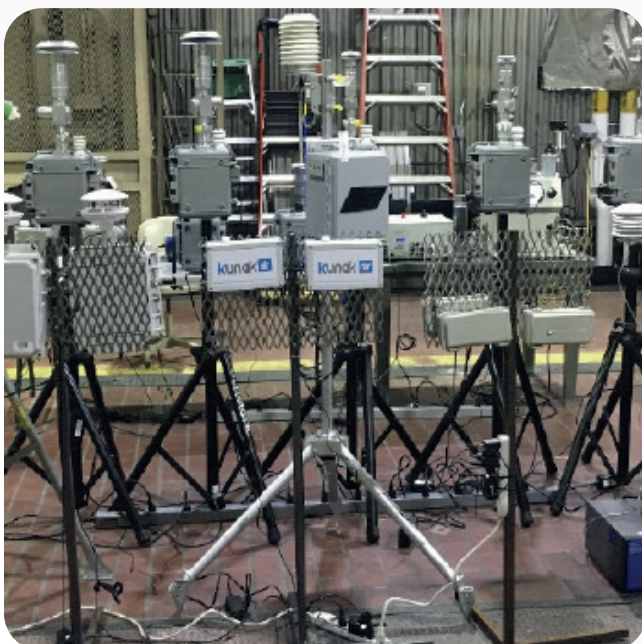
Smoke from wildfires is harmful to health and can irritate the eyes, nose and throat, cause persistent coughing, wheezing and shortness of breath, and worsen heart and lung disease.

The main objectives were:

- Evaluating air quality sensor-based technologies to measure smoke pollutants during wildfires.
- Encourage innovation through the analysis of new air monitoring systems for remote areas.

Measured parameters

- CO (carbon monoxide)
- O₃ (ozone)
- Particulate Matter PM_{2.5}
- Temperature, relative humidity, dew point and atmospheric pressure.





The challenge has raised awareness of the importance of monitoring forest fires, driving the development of sensor technology to protect public health.



Solution

Deployment of the Kunak AIR system to measure particulate matter and pollutants required for the challenge (PM_{2.5}, CO and O₃).

- Air monitoring to detect possible high pollution episodes caused by fires.
- Alert system with detection of pre-established thresholds exceedance.
- Data analysis with advanced graphs such as pollution roses, time variation, calendars...
- Kunak AIR Cloud platform for operational management and data analysis of the monitoring network.
- Embedded wireless communications.

Results

The project underlines the critical importance of early detection of wildfires to protect public health. Smoke from wildfires can have serious health effects, including irritation of the respiratory tract and aggravation of heart and lung diseases.

The need for progress in early detection of wildfires and monitoring of air quality during these events became evident. The implementation of innovative technologies, such as the Kunak AIR system, offered an effective solution to protect public health and manage air pollution in critical situations.

Benefits

Evaluation of the forest fire monitoring solution:

- Maintenance of accuracy during the different conditions carried out during the test.
- Honorable mention awarded to Kunak after obtaining the best score for PM_{2.5}, CO and O₃ without requiring additional calibrations and despite no CO₂ sensor and no 2G cellular coverage in the area.
- Reliable data transmission to solver cloud-based storage & data visualization system during Phase I testing.
- 100% of data was recovered.

